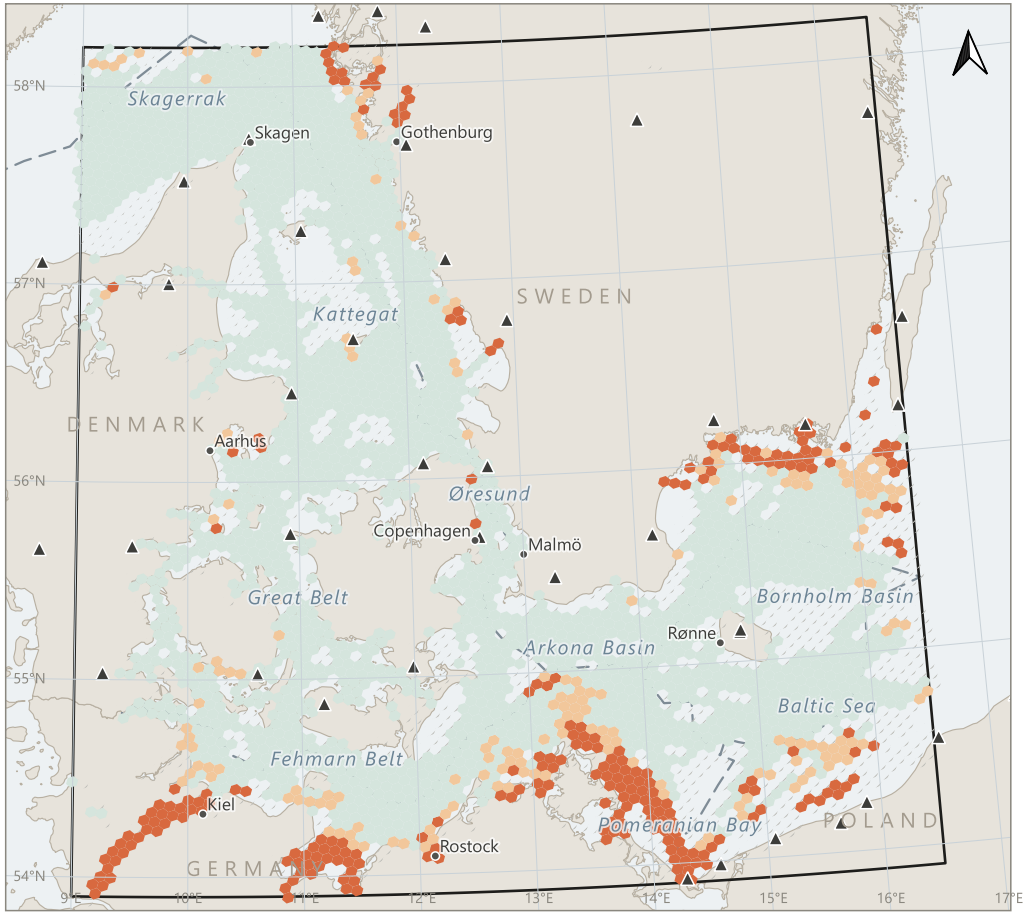
ANALYST VIEW · SHORE-BASED AIS · OPEN DATA

How well the shore network sees, and where to look first

Danish Straits & Bornholm Basin · 13–19 October 2026

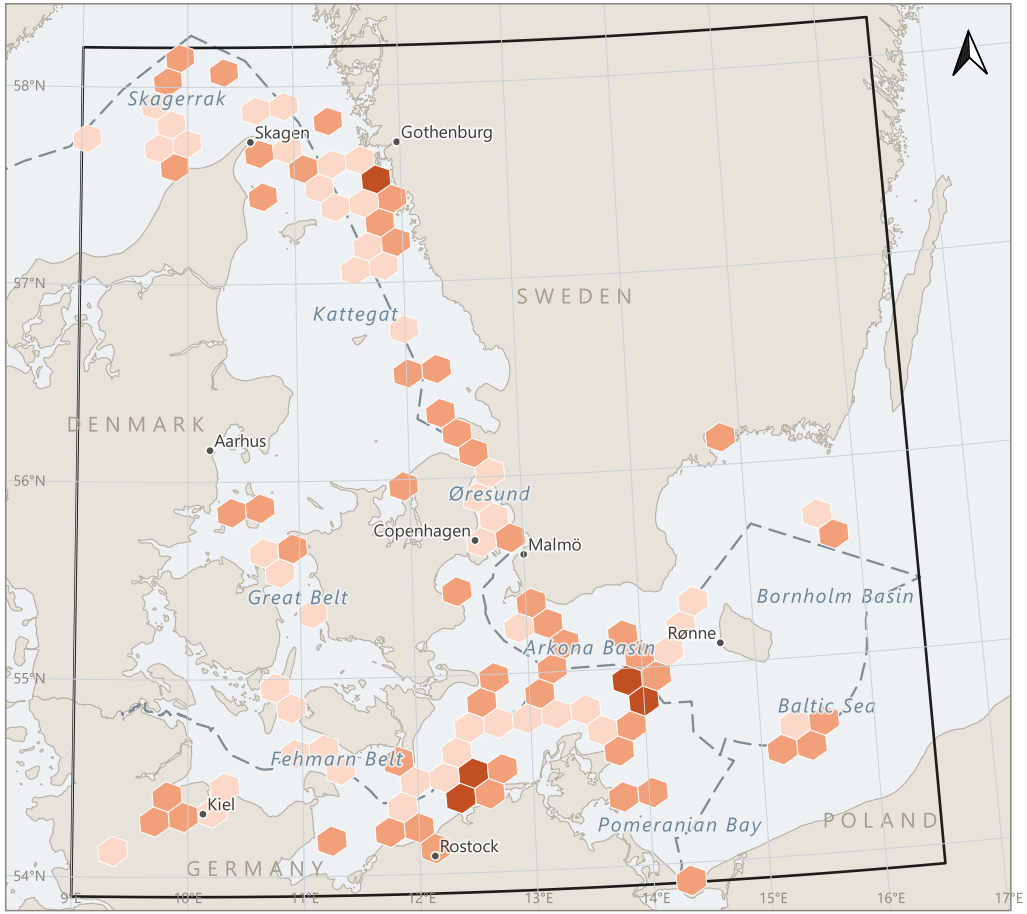


SHORE AIS COVERAGE, MEASURED FROM THE TRAFFIC

Each H3 cell (≈ 36 km²) is rated from the vessels that crossed it during the week: share of 1-minute intervals actually received, and share of expected vessels lost on the way. Counting lost vessels corrects a selection bias: a vessel that is never received leaves no position behind. Triangles: AIS base stations heard by the network.

Shore AIS coverage

- Reliable
- Degraded
- Poor: vessels lost
- Not enough traffic to judge



VIGILANCE INDEX: WHERE TO LOOK FIRST

Mean percentile rank, per H3 cell (≈ 250 km²), of four signals measured as a share of the vessels present: unexplained silences, shared MMSI, slow passages over cables or pipelines, encounters at sea. Cells with fewer than 10 vessels in the week are not rated. A triage tool, not evidence.

Vigilance index

- Low (1 – 15)
- Medium (15 – 30)
- High (> 30)

Data: Danish Maritime Authority (AIS) · Flanders Marine Institute, Marine Regions (EEZ, CC BY 4.0) · Eurostat GISCO (© EuroGeographics) · OpenStreetMap contributors (ODbL). Projection ETRS89-LAEA (EPSG:3035).

Independent portfolio project · Jaurès Daa-Hingbanon, GIS analyst